# The Impact of Daily Exercise on Stress Levels on an Undergraduate Student: A 60-Day Observational Study

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***Abstract*—Stress is a common psychological issue among undergraduate students, often affecting academic performance and mental well-being. This study investigates the relationship between daily physical exercise and stress levels over a 60-day period. A synthetic observational dataset was generated containing exercise activity, sleep duration, workload intensity, and stress ratings. Statistical analysis, including correlation, independent t-tests, and regression modeling, revealed that regular exercise is associated with lower stress levels. These findings suggest that physical activity plays a significant role in stress management among students.**

***Index Terms*—exercise, stress, undergraduate student, mental health, statistical analysis.**

## I. Introduction

### A. Background

Personal data tracking has become increasingly valuable for understanding individual habits, behaviors, and well-being [1]. By monitoring daily activities such as exercise, sleep, and academic workload, individuals can gain insights into behavioral patterns and their effects on mental health [2]. Stress is a common issue among undergraduate students and can negatively affect academic performance and psychological well-being [3]. Understanding behavioral factors that influence stress can help students adopt healthier routines and inform institutional wellness programs [4].

### B. Problem Statement

This study addresses the following research problem: How does daily exercise influence perceived stress levels over time in an undergraduate student, and how do sleep duration and academic workload relate to stress?

### C. Hypotheses

This study evaluates the relationship between daily exercise and perceived stress levels using statistical testing. The following hypotheses were formulated:

**Null Hypothesis ($H_0$):** Daily exercise has no significant effect on perceived stress levels among undergraduate students.

**Alternative Hypothesis ($H_1$):** Daily exercise has a significant effect on perceived stress levels among undergraduate students.

In addition, the study examines the relationships of sleep duration and academic workload with stress levels through correlation and regression analysis.

### D. Objectives

The objectives of this study are as follows:

1) To collect non-sensitive self-tracked daily data on exercise, sleep, workload, and stress.
2) To analyze and visualize temporal patterns and relationships among these variables.
3) To conduct statistical tests to evaluate the significance of observed relationships.
4) To develop a regression model to explain stress levels based on daily behavioral factors.

### E. Scope and Limitations

This study focuses on a single-subject self-tracking dataset collected over a 60-day period. Only non-sensitive variables were recorded. The findings are observational and are not intended to be generalized to the broader student population.

## II. Related Work

Self-tracking and quantified-self research has been widely explored in health and behavioral sciences [1]. Prior studies have examined relationships between physical activity, sleep, and psychological stress using wearable devices and self-reported surveys [5]. Statistical methods such as correlation analysis, t-tests, and regression models are commonly used to analyze behavioral datasets [6]. Recent research has also applied machine learning models for stress prediction based on daily activity data [7]. This study builds on these approaches by applying statistical and modeling techniques to a personal longitudinal dataset.

## III. METHODOLOGY

### A. Participants

The participant in this study was the researcher (the student) who served as the primary subject for data collection and analysis. This self-experimentation approach was adopted to control external variability and to ensure consistency in task execution and system usage during the experiment.

Demographics: The participant is an undergraduate Computer Science student within the typical university age range (18–25 years). The participant has prior experience in computer usage, programming, and interacting with digital systems, which is relevant to the tasks performed in this study. No personally identifiable information was collected or disclosed to maintain privacy and ethical research standards.

### B. Data Collection

Daily data were recorded for 60 consecutive days. The tracked variables included exercise (binary), sleep duration (hours), academic workload intensity (categorical), and perceived stress level (1-5 scale). Data were recorded manually in a spreadsheet and exported as a CSV file for analysis. No sensitive personal information was collected, ensuring ethical compliance.

### C. Operational Definitions

- Day: Numeric, day index (1–60)
- Exercise: Binary (1 = exercised, 0 = no exercise)
- Sleep: Numeric, hours of sleep
- Workload: Categorical (Light, Medium, Heavy)
- Stress: Numeric scale (1–5)

### D. Data Description

Table I presents the variables used in the study, including their type, unit, and description.

TABLE I: Dataset Variables

| Variable | Type | Unit | Description |
|---|---|---|---|
| Day | Numeric | Days | Sequential day index |
| Exercise | Binary | 0/1 | Whether exercise was performed |
| Sleep | Numeric | Hours | Duration of sleep |
| Workload | Categorical | Level | Academic workload intensity |
| Stress | Numeric | Scale (1–10) | Self-reported stress level |

### E. Data Cleaning

Workload categories were encoded numerically (Light = 1, Medium = 2, Heavy = 3). Missing values were checked, and none were observed. No extreme outliers were identified. The variables were used in their original scale without normalization due to the small size of the data set.

### F. Exploratory Data Analysis

Exploratory data analysis included line plots of stress over time, scatter plots of exercise versus stress, and bar charts comparing mean stress levels on exercise and non-exercise days. A correlation heatmap was also generated to visualize relationships among numerical variables.

### G. Statistical Tests

Pearson correlation was used to measure linear relationships between variables. An independent samples t-test was conducted to compare stress levels on exercise versus non-exercise days. These tests were selected due to their suitability for continuous and binary variables and small sample sizes [6]. Statistical significance was evaluated using p-values.

The statistical tests were selected based on the type of variables and research objectives. Pearson correlation was used to assess linear relationships between continuous variables, while the independent samples t-test was used to compare mean stress levels between exercise and non-exercise days. Assumptions of normality and independence were examined through exploratory analysis and diagnostic statistics. Statistical significance was evaluated using p-values.

### H. Modeling and Correlation Analysis

A multiple linear regression model was constructed with stress as the dependent variable and exercise, sleep duration, and workload intensity as independent variables. Model performance was evaluated using R-squared and statistical significance of coefficients.

The regression model assumes linear relationships between predictors and stress, independence of observations, and normally distributed residuals. Model performance was evaluated using the coefficient of determination ($R^2$), which measures the proportion of variance in stress explained by the predictors.

## IV. Results

This section presents the descriptive statistics, visualizations, and statistical test results obtained from the 60-day self-tracked dataset. The findings are reported objectively without interpretation.

### A. Descriptive Statistics

Table II summarizes the mean, median, and standard deviation of the tracked variables.

TABLE II: Descriptive Statistics of Tracked Variables

| Variable | Mean | Median | Standard Deviation |
|---|---|---|---|
| Exercise | 0.566667 | 1.0 | 0.499717 |
| Sleep (hours) | 7.028333 | 7.0 | 0.951554 |
| Workload | 1.900000 | 2.0 | 0.796177 |
| Stress | 3.216667 | 3.0 | 0.958312 |

The descriptive statistics show that exercise was performed on more than half of the observed days (mean = 0.57), indicating moderate consistency in physical activity. Sleep duration averaged approximately 7.03 hours with relatively low variability, suggesting stable sleep patterns throughout the observation period. Workload had an average value of 1.90, indicating that most days involved a moderate academic workload. The mean stress level was 3.22 with moderate variability, showing that perceived stress fluctuated across days but remained within a moderate range overall.

The dataset contains 60 daily observations with numeric and categorical behavioral variables.

## B. Distribution Analysis

Histograms were generated to examine the distribution of stress, sleep duration, and workload.

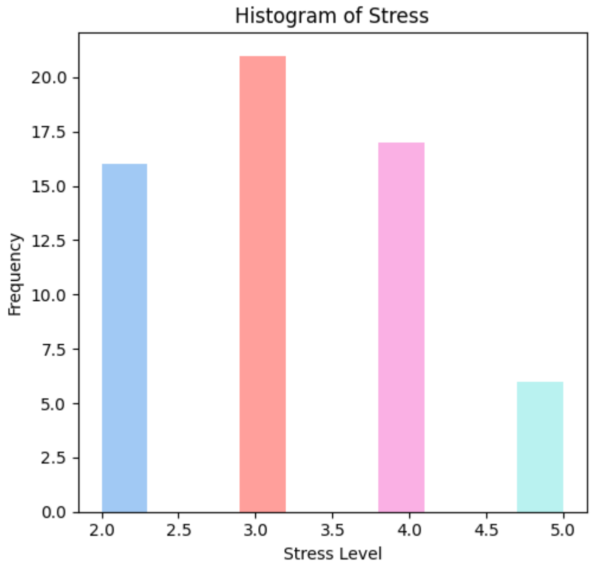


Fig. 1: Distribution of Stress Levels

The stress distribution shows variability across the observation period, with most values concentrated around moderate stress levels. This indicates that perceived stress fluctuated daily without extreme concentration at either very low or very high levels.

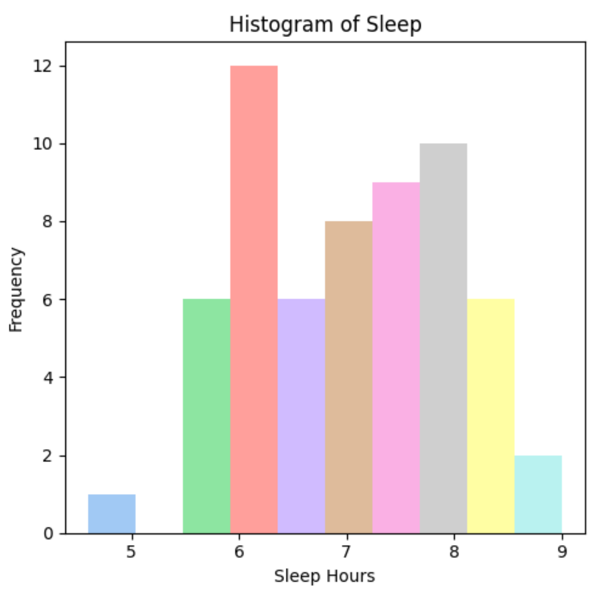


Fig. 2: Distribution of Sleep Duration

The sleep duration distribution is centered around approximately 7 hours, indicating relatively consistent sleep patterns throughout the study period with moderate day-to-day variation.

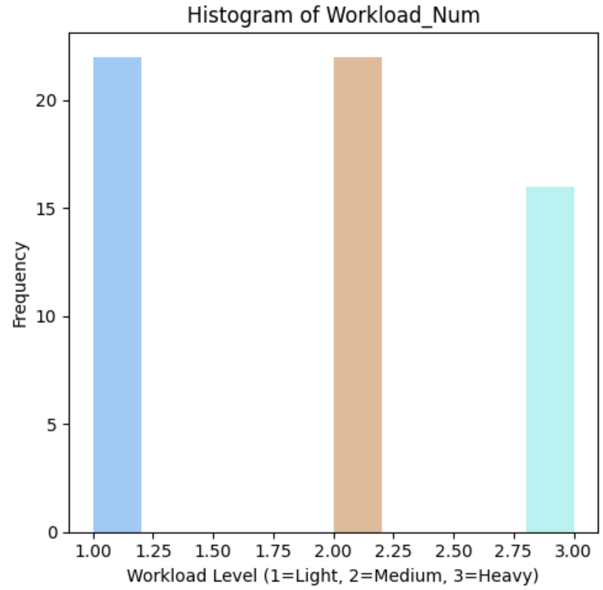


Fig. 3: Distribution of Workload Intensity

The workload distribution shows that most days were categorized as medium workload, with fewer occurrences of light and heavy workloads, reflecting typical academic activity patterns during the observation period.

## C. Time-Series Trends

A time-series plot was generated to observe stress level fluctuations across the 60-day period.

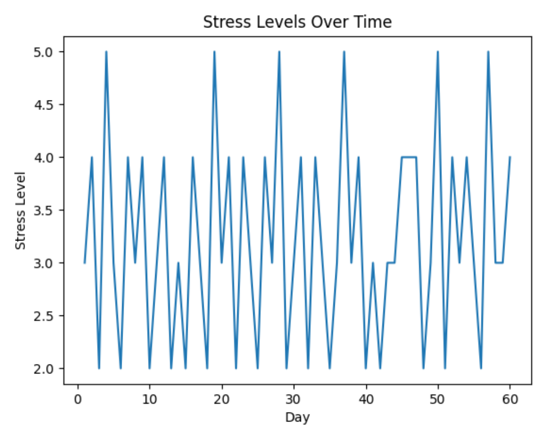


Fig. 4: Stress Levels Over Time

The time-series plot shows fluctuations in stress levels across the 60-day period, reflecting changes in daily exercise behavior, sleep duration, and academic workload over time.

## D. Mean Stress Comparison by Exercise

The bar chart shows that the mean stress level on non-exercise days (Exercise = 0) is noticeably higher than on exercise days (Exercise = 1). This indicates that stress levels were lower on days when exercise was performed.

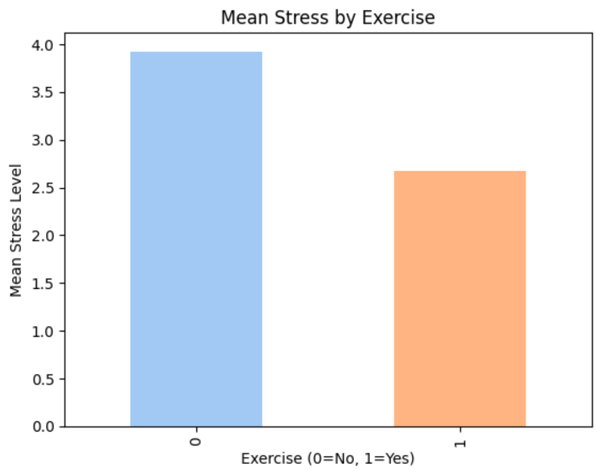


Fig. 5: Mean Stress Levels on Exercise and Non-Exercise Days

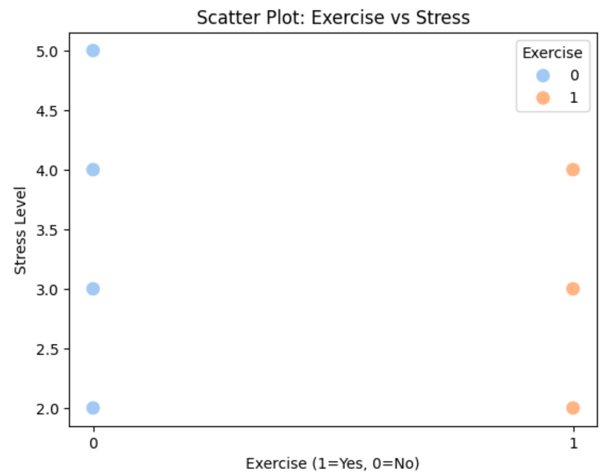


Fig. 7: Scatter Plot of Exercise and Stress Levels

### *E. Correlation Analysis*

The correlation plot shows a moderate negative relationship between exercise and stress and between sleep and stress, indicating that higher exercise frequency and longer sleep are associated with lower stress levels. A strong positive relationship is observed between workload and stress, suggesting that increased academic workload corresponds with higher perceived stress.

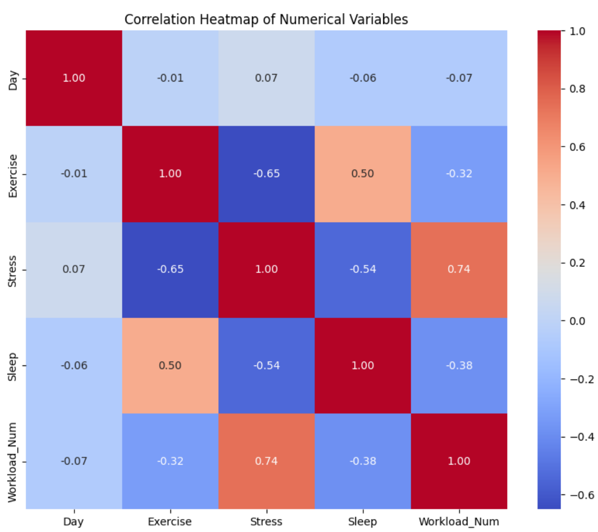


Fig. 6: Correlation Matrix Heatmap

The heatmap highlights the strength and direction of relationships among variables. Workload demonstrates the strongest positive association with stress, while exercise shows the strongest negative association. Sleep exhibits a moderate inverse relationship with stress.

### *F. Scatter Plot*

The scatter plot shows that stress levels tend to be lower on days when exercise is performed compared to days without exercise.

### *G. Statistical Test Results*

An independent samples t-test was conducted to compare stress levels on exercise and non-exercise days. The results indicated a statistically significant difference ($t = -6.51$, $p < 0.001$).

The t-test results indicate a statistically significant difference in stress levels between exercise and non-exercise days, confirming that exercise is associated with lower stress.

### *H. Regression Results*

Multiple linear regression analysis was performed to evaluate the effect of exercise, sleep, and workload on stress levels. The model was statistically significant ($F = 54.99$, $p < 0.001$) and explained 74.7% of the variance in stress levels ($R^2 = 0.747$).

Exercise significantly predicted lower stress levels ($\beta = -0.776$, $p < 0.001$), while workload significantly predicted higher stress levels ($\beta = 0.675$, $p < 0.001$). Sleep was not a statistically significant predictor ($\beta = -0.127$, $p = 0.123$).

Overall, the dataset consisted of daily behavioral measurements collected over 60 days. Descriptive statistics showed variability in sleep, workload, and stress levels. Time-series analysis revealed fluctuations in stress across the observation period. Correlation analysis identified moderate negative relationships between exercise and stress, moderate negative relationships between sleep and stress, and a strong positive relationship between workload and stress. The t-test confirmed significant differences in stress levels between exercise and non-exercise days. Regression modeling identified exercise and workload as significant predictors of stress levels.

### *I. Outlier Analysis*

The following outlier results:

- Omnibus p = 0.014
- Jarque-Bera p = 0.0099
- Skew = 0.605
- Kurtosis = 4.49

Exploratory data analysis did not reveal extreme outliers that required removal. Although minor deviations from normality were observed, all data points were retained as they represented valid real-world measurements and did not significantly distort the regression model.

### J. Hypothesis Testing Summary

The statistical results were used to evaluate the study hypotheses. The independent samples t-test showed a statistically significant difference in stress levels between exercise and non-exercise days ($t = -6.51$, $p < 0.001$). Correlation analysis indicated a moderate negative relationship between exercise and stress ($r = -0.65$). Regression analysis further identified exercise as a significant predictor of stress levels ($\beta = -0.776$, $p < 0.001$).

Based on these results, the null hypothesis ($H_0$) stating that daily exercise has no significant effect on stress levels was rejected, while the alternative hypothesis ($H_1$) was supported.

For the additional variables, workload showed a strong positive relationship with stress and was a statistically significant predictor, while sleep demonstrated a moderate relationship but was not statistically significant in the regression model.

## V. Discussion

This section interprets the findings of the study and explains their implications in relation to the study hypotheses and existing research.

### A. Interpretation of Hypothesis Results

The results of the statistical analyses provide strong evidence regarding the study hypotheses. The null hypothesis ($H_0$), which stated that daily exercise has no significant effect on perceived stress levels, was rejected based on the significant t-test, correlation, and regression findings. The alternative hypothesis ($H_1$) was therefore supported, indicating that exercise significantly influences stress levels.

The negative correlation between exercise and stress and the significant regression coefficient suggest that engaging in daily physical activity is associated with lower perceived stress. These results indicate that exercise plays an important role in managing stress among undergraduate students.

Sleep duration showed a moderate negative relationship with stress; however, it was not statistically significant in the regression model. This suggests that sleep may contribute to stress reduction but does not independently predict stress when exercise and workload are considered simultaneously.

Workload demonstrated the strongest positive relationship with stress and was a statistically significant predictor. This indicates that academic demands are a major contributing factor to increased stress levels.

### B. Comparison to Related Work

The findings of this study are consistent with prior research indicating that physical activity contributes to improved mental well-being and reduced stress levels [5], [4]. Previous studies have also reported that academic workload significantly influences student stress and psychological outcomes [3].

Additionally, research in self-tracking and behavioral analytics supports the use of longitudinal personal data to identify lifestyle factors influencing well-being [1], [2]. However, while some studies found sleep to be a strong predictor of stress, the current study observed that sleep was not statistically significant when analyzed alongside exercise and workload, possibly due to the stronger influence of these factors.

### C. Limitations

This study has several limitations. First, the sample size was limited to a single participant (n = 1), which restricts generalizability. Second, the data were self-reported, which may introduce reporting bias. Third, the duration of data collection was limited to 60 days, which may not capture long-term behavioral trends. Finally, other psychological and environmental factors influencing stress were not included.

### D. Recommendations and Future Work

Future research should involve multiple participants and longer observation periods to improve reliability and generalizability. Automated data collection through wearable devices and mobile applications may reduce self-report bias. Additional behavioral variables, such as screen time, dietary habits, academic deadlines, and social interaction, should also be considered.

Advanced analytical approaches, including machine learning models for stress prediction and clustering of behavioral patterns, may further improve understanding of lifestyle influences on stress [7], [6], [8].

## VI. Conclusion

This study examined the relationship between daily exercise, sleep, workload, and stress using a 60-day self-tracked dataset. The objectives of collecting behavioral data, analyzing patterns, performing statistical tests, and developing a regression model were successfully achieved.

The findings indicate that exercise and workload were the strongest predictors of stress levels, while sleep showed a moderate but non-significant effect when analyzed alongside other variables. Statistical testing confirmed that stress levels differed significantly between exercise and non-exercise days.

These results highlight the value of personal data tracking in understanding behavioral patterns and provide evidence that physical activity plays a meaningful role in stress management among undergraduate students.

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